# Multiscale CNN With Autoencoder Regularization Joint Contextual Attention Network for SAR Image Classification

Zitong Wu, Biao Hou, *Member, IEEE*, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— Synthetic aperture radar (SAR) image classification is a fundamental research direction in image interpretation. With the development of various intelligent technologies, deep learning techniques are gradually being applied to SAR image classification. In this study, a new SAR classification algorithm known as the multiscale convolutional neural network with an autoencoder regularization joint contextual attention network (MCAR-CAN) is proposed. The MCAR-CAN has two branches: the autoencoder regularization branch and the context attention branch. First, autoencoder regularization is used for the reconstruction of the input to regularize the classification in the autoencoder regularization branch. Multiscale input and an asymmetric structure of the autoencoder branch cause the network more to be focused on classification than on reconstruction. Second, the attention mechanism is used to produce an attention map in which each attention weight corresponds to a context correlation in attention branch. The robust features are obtained by the attention mechanism. Finally, the features obtained by the two branches are spliced for classification. In addition, a new training strategy and a postprocessing method are designed to further improve the classification accuracy. Experiments performed on the data from three SAR images demonstrated the effectiveness and robustness of the proposed algorithm.**

***Index Terms*— Attention mechanism, autoencoder regularization, convolutional neural network (CNN), image classification, synthetic aperture radar (SAR).**

## I. Introduction

SYNTHETIC aperture radar (SAR) is an active Earth observation system that can be installed on aircraft, satellites, and other flight platforms. SAR transmits microwaves to the ground at all times and in all weather conditions and receives the backscatter of signal to generate a set of SAR images [1]. SAR, which provides very useful information about terrain and the Earth, in general, has unique advantages in disaster monitoring, environmental monitoring, resource exploration, crop yield mapping, and military applications [2]–[4]. Thus, it has been attracting increased attention from many countries. SAR image classification is a significant issue in SAR image interpretation since each pixel is assigned to a category based on its features. At present, SAR image interpretation technology is lagging behind SAR imaging technology. For many existing SAR image classification systems, their performance relies too heavily on high-quality image labels and important manual features [5]. High-quality labels require a significant amount of manpower and resources, and it is possible to generate errors when marking these labels if researchers do not have enough information about the terrain in the images. Because SAR is imaged with signal backscatter, the images are prone to contamination by large amounts of speckle noise. Speckle noise can greatly affect the quality of the extracted features, resulting in a decline in classification accuracy. Moreover, manual features may not be able to utilize all of the original information. As science and technology have allowed larger images to be collected by SAR systems, the difficulty of classification tasks has increased.

After decades of in-depth study, researchers have proposed many types of classification approaches. In traditional SAR image classification frameworks, a classifier is applied after feature extraction. Numerous feature extraction algorithms have been designed to obtain spatial features from SAR images. The gray-level co-occurrence matrix (GLCM) [6]–[8] can infer statistical relationships from pixel values, such as mean, homogeneity, and variance. Xiang *et al.* [9] proposed a superpixel-generating algorithm based on pixel intensity and location similarity for SAR images. The features and GLCM that are extracted from each superpixel make the image classification results more accurate. The texture descriptors [8], [10] computed from the gray level co-occurrence matrix have been widely utilized to model the intensity and amplitude value of SAR images, such as energy, entropy, and inertia. These statistically based features can globally describe SAR images. Kayabol and Zerubia [11] obtained the texture statistics of SAR images and applied an expectation maximization algorithm to classify the pixels. As the resolution increases, simple texture information cannot adequately describe the complex data structure to meet the needs of classification tasks. In order to obtain better local details of SAR images,

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the Gabor filter [12], and wavelet transform [13] are used to extract their local features. The Gabor filter is sensitive to the edges of the image. It provides valuable direction selection and scale selection to extract texture information in different directions [14]. The wavelet transform is also widely used to capture the texture features with some predefined scales and orientations [15]. All of these methods have their shortcomings, however, in that it is difficult to determine the appropriate number of scales and directions when they are applied to SAR images.

Low-level features, such as textures, may fail in complex environments, so some mid- or high-level feature extraction methods are employed in SAR image classification. The bag-of-visual-words (BoVWs) model combines all of the local features into a global descriptor of a terrain. Li *et al.* [16] used the combination of the BoVW model and the $K$-means clustering algorithm to obtain more types of elementary features. Sparse representation can effectively capture a concise data structure to produce a set of overcomplete dictionaries that can be used to discriminate each sample [17], [18]. Dependent on this property, the sparse representation-based classifiers were proposed for use in a SAR classification task. Hou *et al*. [19] added a double-threshold decision condition to sparse representation, which separates out the reliable pixels from the original SAR images. SAR image pixels are separated layer by layer and ultimately, a sparse representation classifier outputs the classification result. Zhou *et al.* [20] proposed a multilevel reconstruction joint sparse representation algorithm for SAR target detection. Multilevel reconstruction can both suppress background clutter and expand the dataset, while sparse representation computes a training dictionary after reconstruction. Ren *et al.* [21] combined sparse representation, low-rank representation, and manifold graphs to investigate the intrinsic property of PolSAR data. The purpose of this method is to construct a projection matrix from the subspace clustering algorithm to obtain features that facilitate the subsequent classification of images. There are likewise some classification methods based on Bayesian theory, such as conditional random fields. Zhang *et al*. [22] used a conditional random field to describe spectral and spatial information from polarimetric SAR images. Ding *et al*. [23] presented a classification approach for SAR images based on conditional random fields with integration of low-level features and local label context. These methods not only utilize some prior knowledge of SAR image features, but also effectively utilize the statistical characteristics of adjacent regions in the SAR image to improve accuracy. In the SAR classification task, the classifier, after extracting the features, also plays a decisive role. Feature extraction and the classification process of traditional methods are separate [24]. Furthermore, speckle noise increases the difficulty of feature extraction, and it can reduce the effectiveness of traditional methods. Therefore, it is critical to design an end-to-end structure that combines a feature extractor and a classifier.

In recent years, deep learning algorithms and machine learning algorithms have developed rapidly. These algorithms integrate automatically learned effective features and classifiers into a single framework. In general, there are three types of deep learning methods for SAR classification: supervised learning, semi-supervised learning, and unsupervised learning. Many supervised methods have been employed to train labeled samples, such as support vector machine (SVM) [25], random forest (RF) [26], and neural network (NN) [27]. Zhao *et al.* [28] used a deep belief network to extract effective spatial features. The spatial information is also added to improve the final classification performance. Each pixel is represented by a tensor block formed by Pauli values and contextual information [29]. The convolutional neural network (CNN) is then trained to extract the spatial features from the tensor blocks. A process that includes a complex backpropagation algorithm for network training extends the entire network, including both data and parameters, into the complex domain for PolSAR classification [30].

Unlike supervised learning, which requires many labels, unsupervised learning methods, as represented by an autoencoder and a Boltzmann machine, do not require labels. Unsupervised learning can express the essential SAR image data structure, which shows great strengths when there are few labels. Geng *et al.* [31] designed a deep convolutional autoencoder to extract features and perform classification. Wang and Wang [32] proposed a stacked sparse autoencoder with virtual adversarial regularization for PolSAR image classification. This method extracts the nonlocal features by calculating the pairwise similarity of each pixel and its surrounding pixels. In order to address the insufficiency of labels, Qin *et al.* [33] used a Boltzmann machine as a module to constrain adaptive boosting. Liu *et al*. [34] used numerous unlabeled PolSAR pixels to train a new type of restricted Boltzmann machine. An unsupervised generative adversarial network (GAN) algorithm [35] set off a revolution in the field of deep learning. After a GAN is introduced into SAR image classification, the performance of the algorithm is significantly improved. Liu *et al*. [36] proposed a new task-oriented GAN that solved the difficulties in the interpretation of PolSAR images, including PolSAR classification and small-sample problems. Although the semi-supervised learning method compensates for the shortcomings of the supervised method, which requires a large number of labeled samples, it can also perform well when the labeled samples are insufficient. Compared with unsupervised methods, semi-supervised learning can fuse label information to improve classification accuracy. Bi *et al*. [37] modeled the PolSAR image as an undirected graph in which the nodes correspond to the labeled and unlabeled pixels, and designed a graph-based semi-supervised deep learning algorithm. Liu *et al*. [38] proposed a novel spatial-anchor graph to address a small amount of labeled samples and time-consuming issues in large-scale PolSAR terrain classification. Although these algorithms have achieved good results, more effective algorithms are needed to improve the results of SAR image classification.

In this work, a SAR image classification method is proposed, which has been named the multiscale CNN with an autoencoder regularization joint contextual attention network (MCAR-CAN). The MCAR-CAN has two branches: an autoencoder branch and a context attention branch. The first

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branch follows the asymmetrical encoder–decoder structure of a CNN, with a large encoder to extract deep image features and a decoder to reconstruct the original input. The decoder is used for regular encoding, making the entire network more focused on classification rather than reconstruction. The second branch of the MCAR-CAN uses attention mechanisms to extract robust classification features. The classifier is inserted after the autoencoder branch and the contextual attention branch. A new training strategy is then used to learn smoother manifolds. At inference time, the decoder in the autoencoder branch is deprecated. The Markov random field (MRF) is a postprocessing algorithm designed to improve classification accuracy. The contributions of this study are as follows.

1) A SAR image classification algorithm is proposed.
2) A new autoencoder regularization module is proposed for SAR image classification.
3) A new attention module is designed.
4) A new training strategy is used in training.

The rest of this article is arranged as follows. In Section II, our proposed algorithm for SAR image classification is introduced, followed by a short description of the details. Section III presents performance assessments based on a series of simulations. Finally, conclusions are drawn in Section IV.

## II. Methodology

The proposed approach (MCAR-CAN) is based on CNN structure with a two-branch architecture. The first branch is an encoder–decoder structure, with an encoder to extract features and a decoder to reconstruct the image. In the second branch, neural networks focus on spatial context information by utilizing an attention mechanism. We combine the latent vectors from the two branches. The spliced vectors are fed to the classifier and the entire network is trained by using a new strategy. Details of the proposed method are described in this section, and we use bold fonts to represent vectors and matrices.

### A. Architecture

It is well known that classification requires intraclass invariant features which can well represent each class, while reconstruction is required to retain all the information [39]. However, reconstructed information has some effect on classification. In SAR image classification, there is usually a case in which the amount of data is insufficient. To circumvent these issues, the MCAR-CAN is composed of two branches: the autoencoder branch and the spatial context attention branch. The autoencoder can learn good features via regularization when the amount of data is small [40].

CNNs have demonstrated impressive results in various machine vision applications over the past few years. A CNN has a multilayer structure and can automatically learn features [41]. Its shallow convolution layers have a smaller perception domain and learn some local features. Its deeper convolution layers have a larger perception domain and can learn more abstract features. In our method, the autoencoder branch can reconstruct the input image based on an asymmetrical end-to-end structure of a CNN. When the input

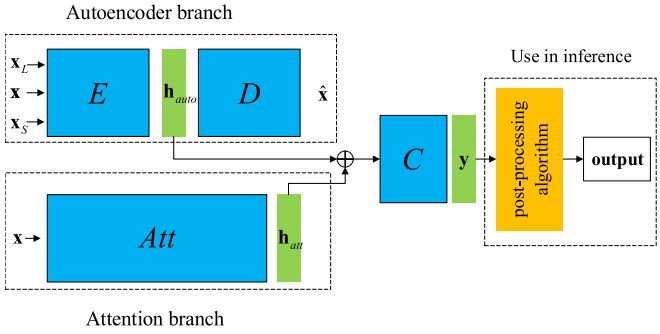



Fig. 1. General description of MCAR-CAN. $E$ and $D$ form an asymmetric autoencoder that is called an autoencoder branch. $Att$ is a neural network that is called the attention branch. $C$ is a classifier. Decoder $D$ is deprecated at inference time. At the end of the inference stage, the post-processing algorithm is used to remove noise points.

consists of multiscale patches, the encoder can determine more multiscale context information, while the decoder produces an approximate image reconstruction. Asymmetric structure can make the autoencoder branch more focused on learning classification information, while the decoder regularizes the encoder. Encoder $E$ takes three multiscale inputs, i.e., $\mathbf{x}_L$, $\mathbf{x}$, and $\mathbf{x}_S$, while it produces a representation $\mathbf{h}_{\text{auto}}$. Also, the decoder $D$ takes $\mathbf{h}_{\text{auto}}$ as an input to generate a reconstruction $\hat{\mathbf{x}}$. It should be noted that the decoder $D$ is deprecated at inference time.

In the spatial context attention branch, the attention mechanism is added to obtain more robust classification features. The input to this branch is $\mathbf{x}$. Since local features at neighboring spatial positions have high relevance, attention branch is helpful in learning spatial dependence and saliency features. The latent representation $\mathbf{h}_{\text{att}}$ can then be obtained. Finally, two latent representations ($\mathbf{h}_{\text{auto}}$, $\mathbf{h}_{\text{att}}$) are concatenated and fed to the fully connected layer to predict a class. The structural sketch, shown in Fig. 1, can be described by the following equations:

$$\begin{aligned} \mathbf{h}_{\text{auto}} &= E(\mathbf{x}_L, \mathbf{x}, \mathbf{x}_S) \quad \hat{\mathbf{x}} = D(\mathbf{h}_{\text{auto}}) \\ \mathbf{h}_{\text{att}} &= \text{Att}(\mathbf{x}) \quad \mathbf{y} = C(\mathbf{h}_{\text{auto}}, \mathbf{h}_{\text{att}}). \end{aligned} \tag{1}$$

This framework can use the refactoring process to facilitate classification when the amount of data is small, since classification requires intraclass invariant features and reconstruction is required to retain all of the information. The encoder loses a lot of information in pooling, while the decoder retains all of the information when reconstructing input. A classifier behind the encoder can produce complementary information which combines two kinds of information from encoder and decoder. Note that the reconstruction $\hat{\mathbf{x}}$ is provided for the discriminator to improve classification results. For the training stage, a new training strategy is proposed. In addition, a postprocessing algorithm is used to further improve classification accuracy. Sections II-B–II-E provide detailed descriptions of each component of the proposed method.

### B. Autoencoder Regularization

The autoencoder branch follows the encoder–decoder-based CNN. The motivation for using the autoencoder structure is to provide additional guidance and regularization to the

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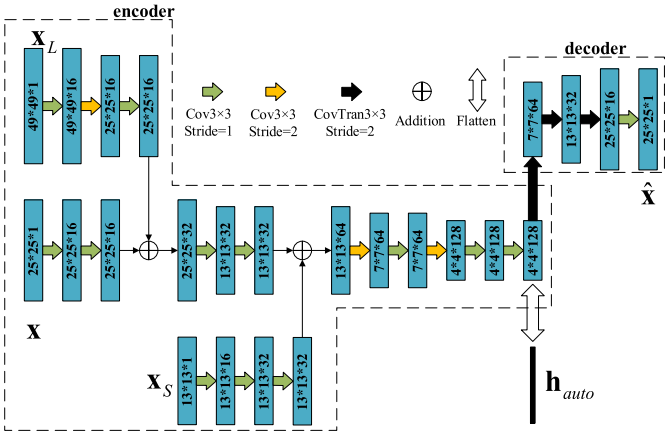



Fig. 2. Architecture of autoencoder branch. Inputs are multiscale SAR patches, followed by 3 × 3 convolution. Multiscale patches transform into 128 feature maps with size 4 × 4 after multiple convolutions. The decoder reconstructs the feature maps into an input image, the decoder is used only during training to regularize the shared encoder.

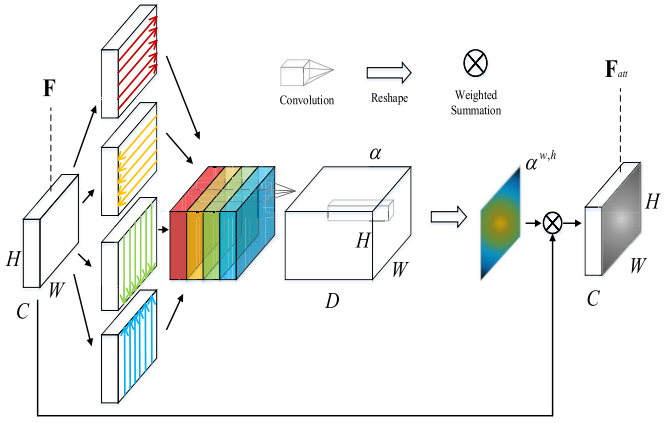



Fig. 3. Architecture of the attention module.

encoder component, since the training dataset size is limited. A schematic of the autoencoder branch of the MCAR-CAN is shown in Fig. 2.

The context information is important for obtaining the category of each pixel in the SAR image. Thus, multiscale SAR patches centered on the pixel are used as input. $\mathbf{x}_L$ is the patch that has the larger size, with a length and width of 49 pixels each. Three convolution layers transform the input $\mathbf{x}_L$ into 16 feature maps, each with a size of 25 × 25. $\mathbf{x}$ and the reconstruction $\hat{\mathbf{x}}$ are the same size (25 × 25). Feature maps obtained from $\mathbf{x}_S$ with a size of 13 × 13 contain the relationship information between a classified pixel and adjacent pixels. In these three convolutional components, context information of multiscale patches is obtained and merged. This merger by addition in pixel space is analogous to signal superposition in which the low-pass signal and high-pass signal are summed. After merging using the addition operation, the feature maps are encoded by four convolution layers. Encoder $E$ then obtains 128 feature maps the size of which is 4 × 4. Decoder $D$ is behind encoder $E$, where the structure of the decoder $D$ is dissymmetric to encoder $E$. The reconstruction $\hat{\mathbf{x}}$, which is an approximation of $\mathbf{x}$, is obtained after four transposed convolution layers. Feature maps obtained by encoder $E$ can be flattened to gain latent representation $\mathbf{h}_{\text{auto}}$. This branch is composed of convolutional layers with a padding of 1, each with an activation function RELU; the convolution kernel size is 3 × 3.

### C. Attention Mechanism With CNN for Spatial Context

The attention mechanism is based on a reasonable assumption that human vision does not process an entire image at once, and it only focuses on specific parts of the entire visual space at "high resolution" while perceiving the surrounding in "low resolution" [42], [43]. The context attention branch of the MCAR-CAN attempts to extract spatial dependence and saliency features. The attention mechanism [44]–[46] is added onto the CNN in order to focus the relationship between vicinity and significant features on the entire input patch. Cui *et al.* [47] used the attention module to extract the rich semantic information of SAR images for ship detection. Mei *et al.* [48] used spectral and spatial information to propose a spectral–spatial attention network for hyperspectral image classification. Liu and Han [49] proposed PiCANet to generate attention over the context regions for each pixel. Focusing on the aerial view characteristics of SAR images, we improved PiCANet to obtain better classification features. Four long short-term memory (LSTM) networks replaced the original bidirectional LSTM networks. $\mathbf{F} \in \mathbb{R}^{W\times H\times C}$ is defined as the convolution feature map, where $W$, $H$, and $C$ represent its width, height, and number of channels, respectively. For the attention module, the network framework is illustrated in Fig. 3.

Here, an effective residual model is employed as a basic module, which uses four LSTM networks to scan feature maps in the horizontal and vertical directions. Each column of $\mathbf{F}$ or each row of $\mathbf{F}$ is fed into these four LSTMs, and the hidden state of each pixel can then remember the context information around each pixel. By scanning in four directions, the context can be mixed together to transmit information between pixels. We tend to make each pixel to "see" the overall feature map and add the weight generated from each pixel to get the final attention weight. Thus, a convolution layer is used to transform the feature map to $D$ channels, where $D = W \times H$. In addition, $\mathbf{p}^{w,h} \in \mathbb{R}^D$ is defined as a feature vector of pixel at the location $(w, h)$. It is normalized by a softmax function to obtain an attention weight $\boldsymbol{\alpha}^{w,h} \in \mathbb{R}^D$

$$\alpha_i^{w,h} = \frac{\exp\left(p_i^{w,h}\right)}{\sum_{j=1}^{D}\exp\left(p_j^{w,h}\right)} \tag{2}$$

where $i \in \{1, \ldots, D\}$, and $\alpha_i^{w,h}$ is the context relation between the pixel at the $i$th location $(w, h)$ and the overall feature map. Eventually, for pixel $(w, h)$, the attended contextual feature $\mathbf{F}_{\text{att}}$ with the same size of $\mathbf{F}$ can be obtained by the following equation:

$$\mathbf{F}_{\text{att}}^{w,h} = \sum_{i=1}^{D} \alpha_i^{w,h}\mathbf{f}^{w,h} \tag{3}$$

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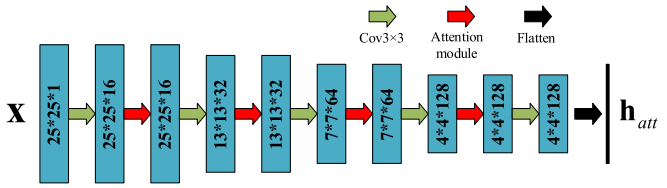



Fig. 4. Framework of attention branch. The convolution and attention operations alternate, extracting features from input.

where $\mathbf{f}^{w,h} \in \mathbb{R}^C$ is the convolutional feature at the location $(w, h)$ in $\mathbf{F}$, $\mathbf{F}_{\text{att}}^{w,h}$ is the feature at the location $(w, h)$ in $\mathbf{F}_{\text{att}}$. In addition, it can improve computational efficiency for all pixels for which (3) is implemented by convolution. As Fig. 4 shows, the attention branch of the MCAR-CAN that is based on the attention module is revealed.

The input of this branch is the SAR patch $\mathbf{x}$. This branch consists of five convolutional layers and four attention modules, which are alternated. The padding of the convolutional layer is 1, and the activation function is RELU. In the attention module, all of the LSTMs have 32 hidden layers. Finally, the latent representation $\mathbf{h}_{\text{att}}$ is obtained by flattening.

### D. Training Strategy

*1) Traditional Training Strategy:* The latent vector $\mathbf{h}_{\text{auto}}$, which forms an autoencoder branch, and the latent vector $\mathbf{h}_{\text{att}}$, which forms an attention branch, are spliced to produce a data representation $\mathbf{h}$. Data representation $\mathbf{h}$ is fed into classifier which consists of three fully connected layers and a softmax layer to generate the classification result $\hat{\mathbf{y}}$. And the label is $\mathbf{y}$. The size of all the convolutional kernels in the MCAR-CAN is $3 \times 3$. The activation function of the convolution layer and the transposition convolution is RELU, and the activation function of the full-connected layer is LeakyRELU. In traditional training strategy, the loss function consists of two terms

$$L = \lambda_C L_C + \lambda_R L_{\text{REC}}. \tag{4}$$

The classification loss $L_C$ is a cross-entropy term

$$L_C = -\sum_k \left[ y^{(k)} \log \hat{y}^{(k)} + \left(1 - y^{(k)}\right) \log\left(1 - \hat{y}^{(k)}\right) \right] \tag{5}$$

where $k$ represents the number of classes. The MCAR-CAN captures different levels of context information and generates reconstructions of $\mathbf{x}$ patches. $L_{\text{REC}}$ is the reconstruction loss, which is a simple mean-squared error between $\mathbf{x}$ and $\hat{\mathbf{x}}$

$$L_{\text{REC}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2. \tag{6}$$

Each term is weighted by a corresponding parameter $\lambda$.

*2) Mixed Training:* Interpolation has been frequently reported as a qualitative experimental result in studies on autoencoding [50]–[52] and latent-variable generative models [53], [54] in general. By mixing latent variables, a neural network can produce a combination of semantically meaningful corresponding data points. The ability to interpolate can be useful, allowing the neural network to learn the potential space with a definite structure in order to promote better classification. If data are inserted into the potential space between two points, the data space will produce smooth semantic distortion, meaning that data that are more similar in the potential space are semantically similar. This property may suggest that semantically similar points may be clustered together in latent space. If there is a discontinuity in the data space, this may result in the representation being less functional [55]. This contact between interpolation and a data manifold has been investigated in the context of representation learning [56] and regularization [57]. Taking into account this situation, Berthelot *et al.* [55] proposed a self-encoding framework for reconstruction tasks and demonstrated the validity of this theory. A new training strategy is also based on this idea, namely, that of inserting mixed data representations to learn smoother manifolds. Fig. 5 illustrates this training process. Suppose that $\mathbf{h}_1$ and $\mathbf{h}_2$ are representations of two data points ($\mathbf{x}_1$, $\mathbf{x}_2$), which are labeled $\mathbf{y}_1$ and $\mathbf{y}_2$. Usually, latent representation is combined using a convex combination so that interpolation is equivalent to calculating $\mathbf{h}_{\text{mix}} = \beta \mathbf{h}_1 + (1 - \beta)\mathbf{h}_2$, where $\beta \in [0, 1]$. Here, the mixture interpolation $\mathbf{h}_{\text{mix}}$ is labeled $\mathbf{y}_{\text{mix}} = \beta \mathbf{y}_1 + (1 - \beta)\mathbf{y}_2$. $\mathbf{h}_{\text{mix}}$ is then input into the classifier to obtain the output $\hat{\mathbf{y}}_{\text{mix}}$.

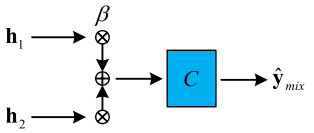



Fig. 5. Schematic of mixed training.

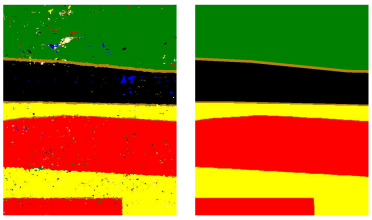



Fig. 6. Before and after using a postprocessing algorithm. (a) No postprocessing algorithm. (b) Postprocessing algorithm processing result.

In mixed training strategy, the loss function consists of three terms

$$L = \lambda_C L_C + \lambda_{R1} L_{\text{REC1}} + \lambda_{R2} L_{\text{REC2}}. \tag{7}$$

The classification loss $L_C$ is a cross-entropy term

$$L_C = -\sum_k \left[ y_{\text{mix}}^{(k)} \log \hat{y}_{\text{mix}}^{(k)} + \left(1 - y_{\text{mix}}^{(k)}\right) \log\left(1 - \hat{y}_{\text{mix}}^{(k)}\right) \right] \tag{8}$$

where $\hat{\mathbf{y}}_{\text{mix}}$ represents the output results, $\mathbf{y}_{\text{mix}}$ represents the labels. $L_{\text{REC1}}$ and $L_{\text{REC2}}$ are the reconstruction loss

$$L_{\text{REC1}} = \|\mathbf{x}_1 - \hat{\mathbf{x}}_1\|_2^2 \quad L_{\text{REC}} = \|\mathbf{x}_2 - \hat{\mathbf{x}}_2\|_2^2. \tag{9}$$

### E. Postprocessing Algorithm

Because each pixel is independent, there is no relationship between the classification results of adjacent pixels [58]. At pixel-by-pixel inference time, the result of the classification may show speckles; i.e., the classification result of a pixel is significantly different from the surrounding pixels, such as the case shown in Fig. 6(a). In order to overcome these classification errors, the MRF [59] is introduced as a postprocessing

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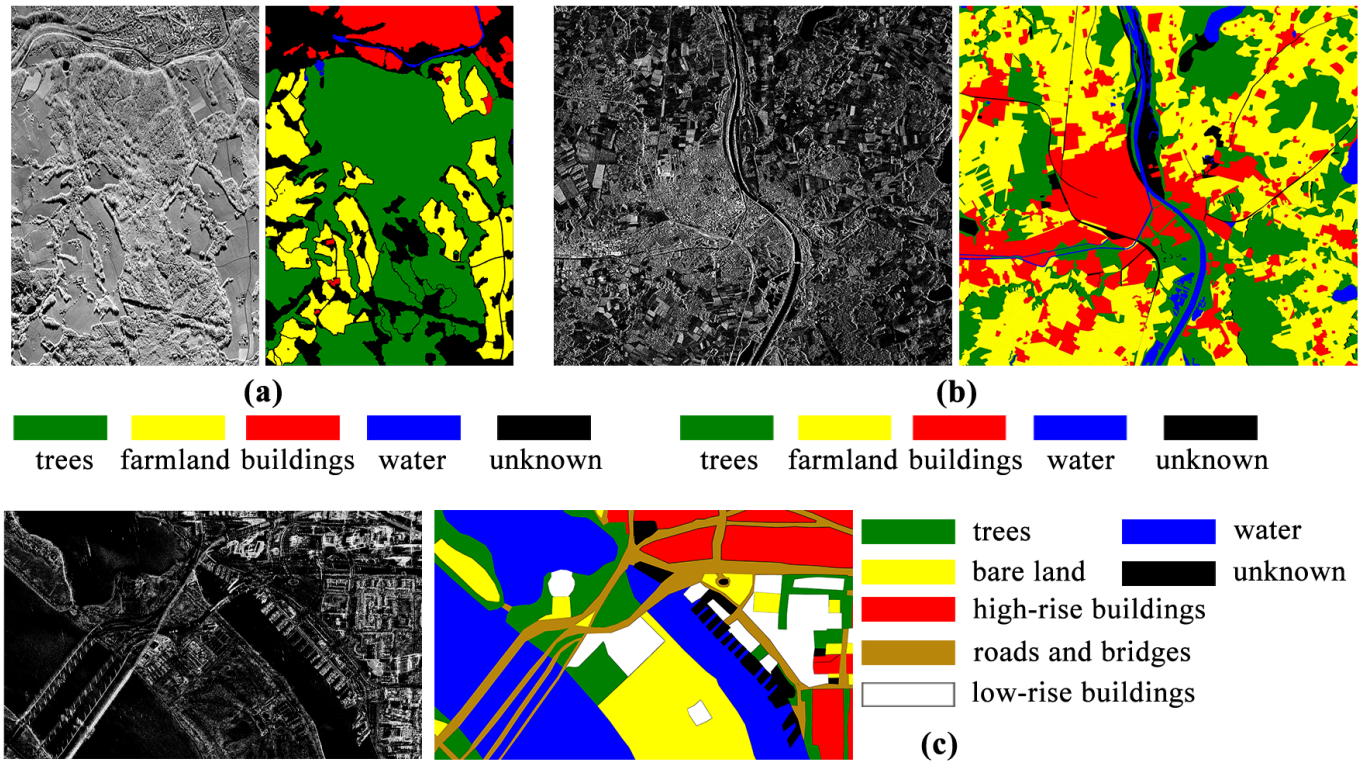



Fig. 7. SAR data adopted in the experiments. (a) Bavaria SAR image. (b) Rosenheim SAR image. (c) Washington SAR image.

algorithm to improve regional connectivity. The MRF is a type of stochastic process, and its original model is a Markov chain. Its main feature is that, under the condition that the current state (now) is known, its future changes (forthcoming) are not dependent on past changes, but only on the current state. The classification result generated by the MCAR-CAN can be used as the initial state of the MRF. Each pixel is affected by pixels in its $3 \times 3$ neighborhood, called clique $G$. The joint probability distribution of the classification result $y$ is

$$p(y) = (\exp\{-U(y)/T\})/Z \tag{10}$$

where $T$ is a constant, $U(y) = \sum_{g \in G} V_g(y)$ is a potential function, $V_g(y)$ is potential energy, and $Z$ is the normalization coefficient. According to the Bayesian formula, the estimated value of $y$ needs to maximize the posterior probability

$$y = \arg \max_{y \in \{1,\ldots,K\}} \{p(y|x)\} = \arg \max_{y \in \{1,\ldots,K\}} \{p(x|y)p(y)\} \tag{11}$$

where $K$ is the number of classes. As shown in Fig. 6(b), the accuracy of the classification results is further improved after optimization of the postprocessing algorithm.

## III. Experiments

### A. Experimental Data

To determine the validity of the MCAR-CAN, we performed experiments on three SAR images and compared the MCAR-CAN with other algorithms. We also explored the contribution of each module by running classification experiments with and without modules, and gradually adding more complexity to the architecture. The first SAR image was acquired in Bavaria, Germany with an HH-polarization mode, and is shown in Fig. 7(a). It has 6187 $\times$ 4278 pixels and its image resolution is 1 m $\times$ 0.6 m. There are five terrains in the Bavaria image: buildings, water, trees, farmland, and unknown. The second SAR image, which was obtained from Rosenheim, Germany, with an HH-polarization mode, is shown in Fig. 7(b). Its size is 6030 $\times$ 6557 pixels and its resolution is 1 m. There are five terrains in the Rosenheim image: buildings, water, trees, farmland, and unknown. The last SAR image was obtained from Washington, DC, USA with an HH-polarization mode; see Fig. 7(c). The size of the Washington image is 4489 $\times$ 7579 pixels and its resolution is 0.3 m. In the Washington image, seven terrains are marked: water, trees, high-rise buildings, low-rise buildings, roads and bridges, bare land, and unknown. To the right of each original SAR image is its corresponding ground truth. All of the experiments in this study were implemented in Python3.6 on a workstation with an Intel Xeon E-2186M CPU, Quadro P5200 GPU, and 64 GB of memory.

### B. Contributions of Modules

Since the architecture of MCAR-CAN is complex, we explored the contributions of network modules by running classification experiments with and without modules. Training strategy and postprocessing method have an impact on the experimental results, so we also evaluated the effects of them.

*1) Networks:* The MCAR-CAN is composed of two branches: the autoencoder branch and the spatial context

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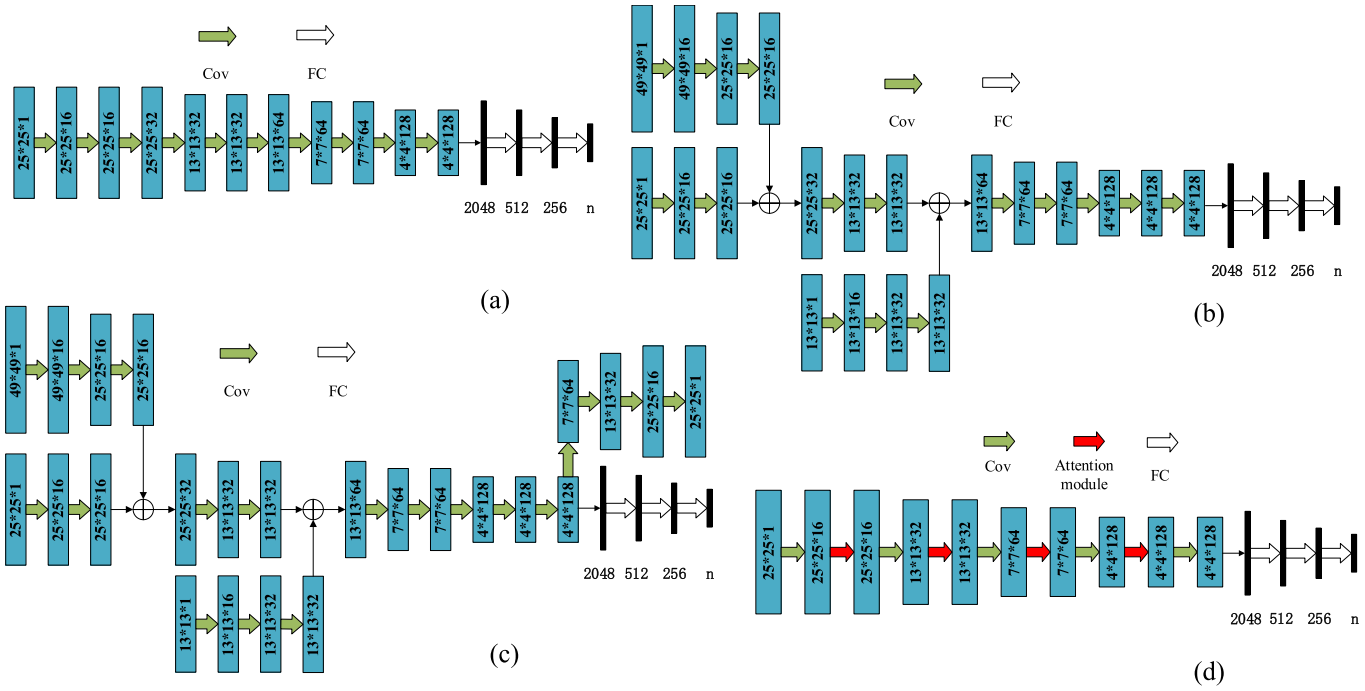



Fig. 8. Architecture of networks. (a) Baseline network. (b) Baseline network with multiscale input. (c) Autoencoder branch. (d) Attention branch.

attention branch. And the autoencoder branch contains multiscale input and autoencoder regularization. Thus, we take the main structure of autoencoder branch as a baseline network, as shown in Fig. 8(a). Multiscale input, autoencoder regularization, and attention branch were gradually added to baseline. And they finally formed a complete MACR-CAN. The baseline network with multiscale input (Baseline-MS) is shown in Fig. 8(b). The autoencoder branch is shown in Fig. 8(c). Three fully connected layers were spliced after attention branch in Fig. 8(d). Baseline network, Baseline-MS, autoencoder branch, attention branch, and MCAR-CAN were utilized to classify three SAR images.

Autoencoder branch network and MCAR-CAN contained autoencoder regularization module, so these two networks were trained according to formula (4). We did an experiment to determine the parameters in Rosenheim SAR image. 10% of the pixels were randomly selected as the training set. Fixed the parameter $\lambda_C$ to 1, we tried to change the value of $\lambda_R$ to train the networks. The experimental results with different $\lambda_R$ are shown in Fig. 9. When $\lambda_R$ equals 0.3, autoencoder branch network and MCAR-CAN can obtain the best classification results. Thus, $\lambda_C$ is equal 1, and $\lambda_R$ is 0.3 in later experiments.

In order to explore the performance of each module, Baseline network, Baseline-MS, autoencoder branch network, attention branch network, and MCAR-CAN were trained in three SAR images. 10% of the pixels were randomly selected as the training set in each SAR image. Autoencoder branch network and MCAR-CAN were trained according to formula (4). And a cross-entropy loss of outputs and labels was used to train Baseline, Baseline-MS, and attention branch network. The experimental results are demonstrated in Table I. For the convenience of representation, we used Image 1

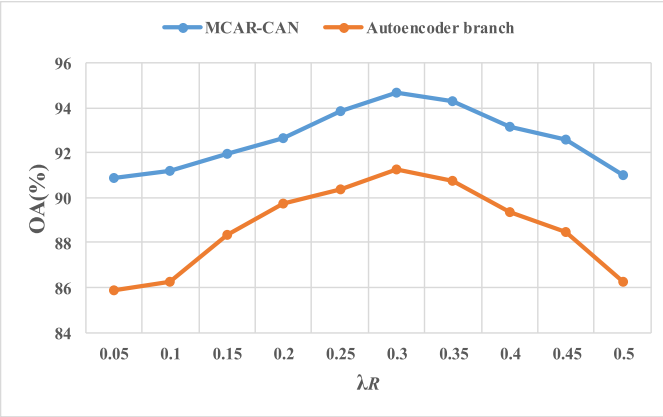



Fig. 9. Overall accuracy of results with different $\lambda_R$.

TABLE I
OA OF CLASSIFICATION RESULTS ON THREE SAR IMAGES (%)

| | Image 1 | Image 2 | Image 3 |
|---|---|---|---|
| Baseline | 80.62 | 81.98 | 61.71 |
| Baseline-MS | 87.32 | 86.27 | 81.09 |
| autoencoder branch | 90.77 | 91.25 | 88.56 |
| attention branch | 89.61 | 90.96 | 85.40 |
| MCAR-CAN | 95.61 | 94.68 | 94.36 |

to represent the Bavaria SAR image, Image 2 to represent the Rosenheim SAR image, and Image 3 to represent the Washington SAR image.

From Table I, it is easy to find that the classification performance of Baseline is the worst one. The accuracy of Baseline is about 80% in the first two SAR images. In the third SAR image, the classification performance of Baseline is greatly reduced. Compared with Baseline, multiscale inputs are added to Baseline-MS. Multiscale patches provide

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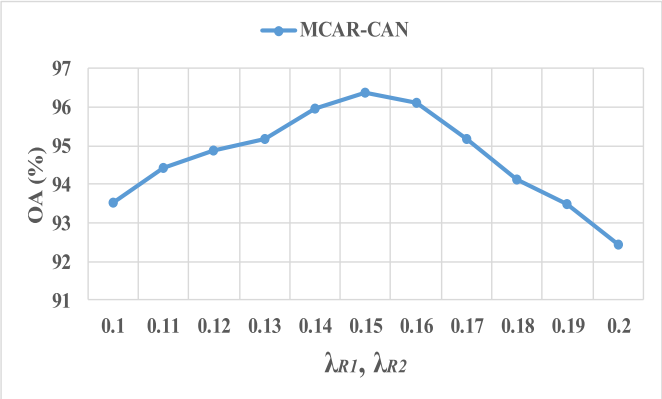



Fig. 10. Overall accuracy of results with different $\lambda_{R1}$ and $\lambda_{R2}$.

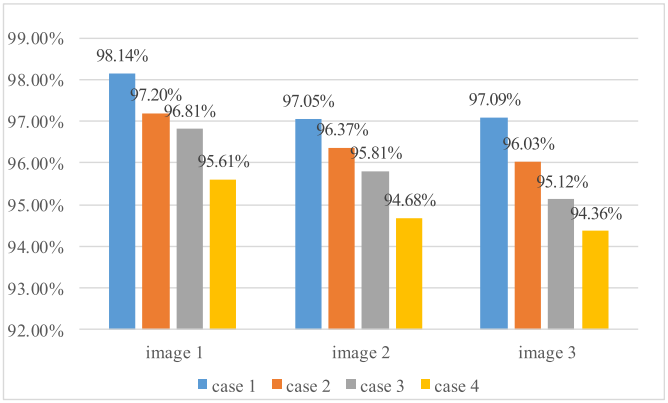



Fig. 11. Overall accuracy of four cases.

different neighborhood information for the classified pixels. Baseline-MS can learn more robust features from multiscale patches. The classification accuracy of Baseline-MS is much higher than that of Baseline. Especially in the third image, the classification accuracy is improved from 61.71 to 81.09. Baseline-MS adds autoencoder regularization to autoencoder branch. Comparing the experimental results of Baseline-MS and autoencoder branch, we can get that the autoencoder regularization can improve the classification accuracy by 3%–7%. Attention branch are trade on learning features from spatial information and it can grasp spatial features between neighboring pixels by adding attention weights. These saliency features from attention branch can better describe the significant area. The classification accuracy of attention branch is higher than 85% in three SAR images. The complete MCAR-CAN is a combination of autoencoder branch and attention branch. The features which are generated from autoencoder branch and the saliency features which are generated from attention branch complement each other to improve classification performance. The classification accuracy of MCAR-CAN is higher than 94% in three SAR image experiments.

*2) Mixed Training and Postprocessing Algorithm:* In order to determine the parameters for proposed strategy, we trained MCAR-CAN according to formula (7). $L_{\mathrm{REC1}}$ and $L_{\mathrm{REC2}}$ had the same effect on the formula (7), so the values of $\lambda_{R1}$ and $\lambda_{R2}$ were equal. By fixing the parameter $\lambda_C$ to 1, we tried to change the value of $\lambda_{R1}$ and $\lambda_{R2}$ to train the MCAR-CAN. The classification results with different parameters are shown in Fig. 10. When $\lambda_{R1}$ and $\lambda_{R2}$ equal 0.15, the MCAR-CAN can obtain the best results. In the subsequent experiments, the values of $\lambda_C$, $\lambda_{R1}$, and $\lambda_{R2}$ were 1, 0.15, and 0.15, respectively.

We evaluated the effects of mixed training strategy and postprocessing method (the MRF) on the MCAR-CAN. More specifically, the mixed training strategy and postprocessing method were combined with the original MCAR-CAN one by one. This resulted in four cases for our SAR experiment: MCAR-CAN with mixed training and with MRF (Case 1), MCAR-CAN with mixed training and without MRF (Case 2), MCAR-CAN with MRF and without mixed training (Case 3), and MCAR-CAN without mixed training and without MRF (Case 4). Fig. 11 depicts the overall accuracy (OA) of the experimental results in these four cases. It is obvious that the OA of Case 1 was the highest, and the OA of Case 4 was the lowest. Case 2 averaged 0.8%–1.2% lower than Case 1, but 0.2%–0.9% higher than Case 3. There was a 2%–3.5% gap between Cases 1 and 4. The mixed training strategy and the MRF were effective in that they helped the MCAR-CAN improve its accuracy by 2%–3.5%. The higher accuracy of Case 2 compared with Case 3 proves that mixed training can promote the accuracy of the entire algorithm more than the MRF.

### C. Classification Comparison and Analysis

In order to demonstrate the performance of our algorithm, we compared the MCAR-CAN with four other methods: the CNN, dual path network (DPN) [60], neural architecture search network (NasNet) [61], and residual network (ResNet) [62]. The mixed training strategy was used to train MCAR-CAN, while the other four comparison methods were trained using a cross-entropy loss. We also used postprocessing algorithms to optimize the results of the comparison algorithm. The CNN comparison algorithm consists of seven convolutional layers and three fully connected layers, in which the convolutional layers with a step size of 2 replace the pooling layers. DPN which was proposed by Chen *et al.* [60] is a combination of ResNet and dense network. In our experiments, DPN network consists of eleven DPN units. NasNet which consists of reduction cells and normal cells can do architecture search with reinforcement learning. In our experiments, NasNet consists of sixteen normal cells and three reduction cells. ResNet which was proposed by He *et al.* [62] has powerful performance in image classification tasks. In our experiments, ResNet consists of six residual units. The architectures of four comparison methods are shown in Table II. All of the comparison algorithms used 25 × 25 SAR patches as input data. In each SAR image, 10% of the pixels were randomly selected as the training set. And three evaluation indicators—OA, average accuracy (AA), and kappa coefficient—were used to measure classification accuracy.

The classification visual results of the three SAR images are shown separately in Figs. 12–14. Fig. 12 reveals that the MCAR-CAN can achieve a better result than the comparison

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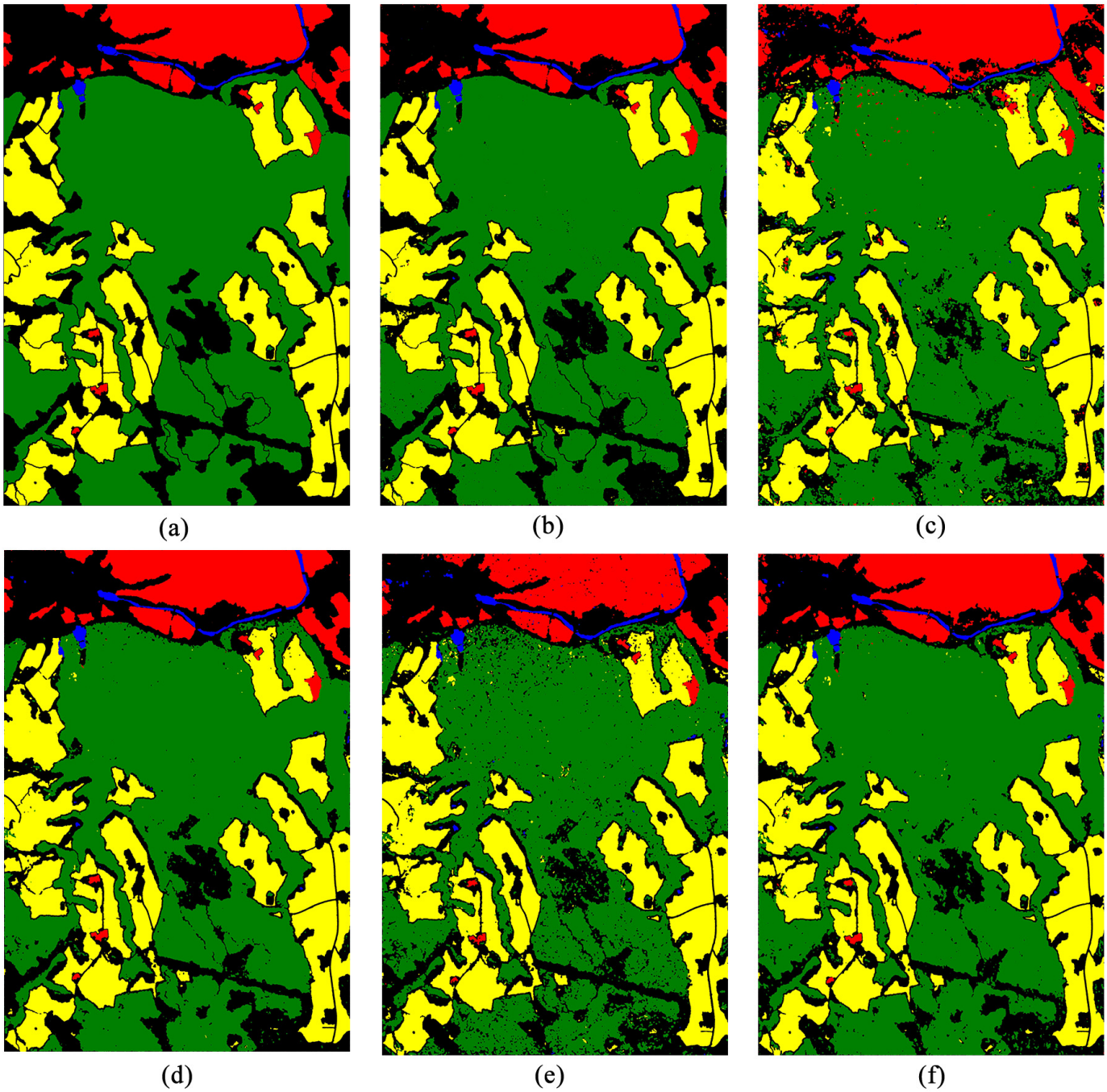



Fig. 12. Classification maps of each method on Bavaria SAR image. (a) Ground truth. (b) MCAR-CAN. (c) CNN. (d) DPN. (e) NasNet. (f) ResNet.

algorithms in the Bavaria SAR image. Only a few pixels were classified incorrectly. Benefiting from the strong features extracted from the proposed method, the classification result of the MCAR-CAN [Fig. 12(b)] was smoother than the results of the other methods. The outline of each area was well extracted. It is worth noting that we used a postprocessing method to improve the connectivity of the classification map. Still, our proposed method was more effective. In the result of CNN [Fig. 12(c)], the texture of the unknown was cluttered. Many pixels of unknown are classified into other classes. The CNN algorithms cannot extract smooth contours for the unknown categories, since they are not robust for speckle noise. And numerous water pixels were classified as other pixels, resulting in the production of an incomplete water area. In the result of NasNet [Fig. 12(e)], a lot of spots in the building area and tree area. These spots were the pixels that are misclassified by other classes to unknown. And many classification errors occurred in the region of buildings, water, trees, and unknown areas. Although the classification accuracy was slightly worse, CNN and NasNet results could also clearly show the distribution of various terrains in the Bavaria image. At the same time, the experimental results of DPN [Fig. 12(d)] and ResNet [Fig. 12(f)] were very similar to the ground truth. Only a small number of pixels are misclassified.

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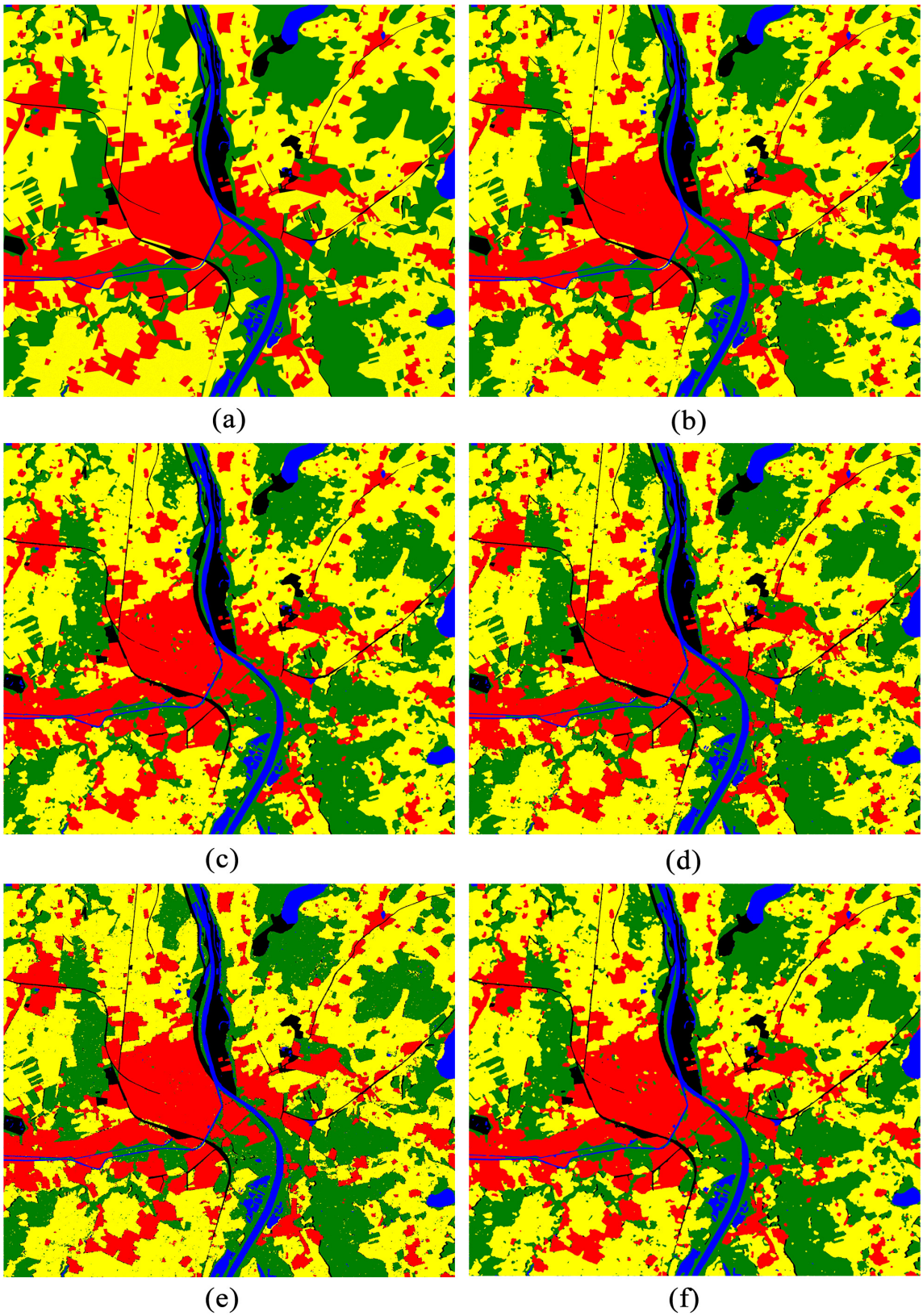



Fig. 13. Classification maps of each method on the Rosenheim SAR image. (a) Ground truth. (b) MCAR-CAN. (c) CNN. (d) DPN. (e) NasNet. (f) ResNet.

However, the latter quantitative analysis shows that the classification result of MCAR-CAN is better than that of DPN and ResNet.

The Rosenheim SAR image contains five types of terrain (buildings, water, trees, farmland, and unknown). Fig. 13 displays the experimental results of the MCAR-CAN [Fig. 13(b)]

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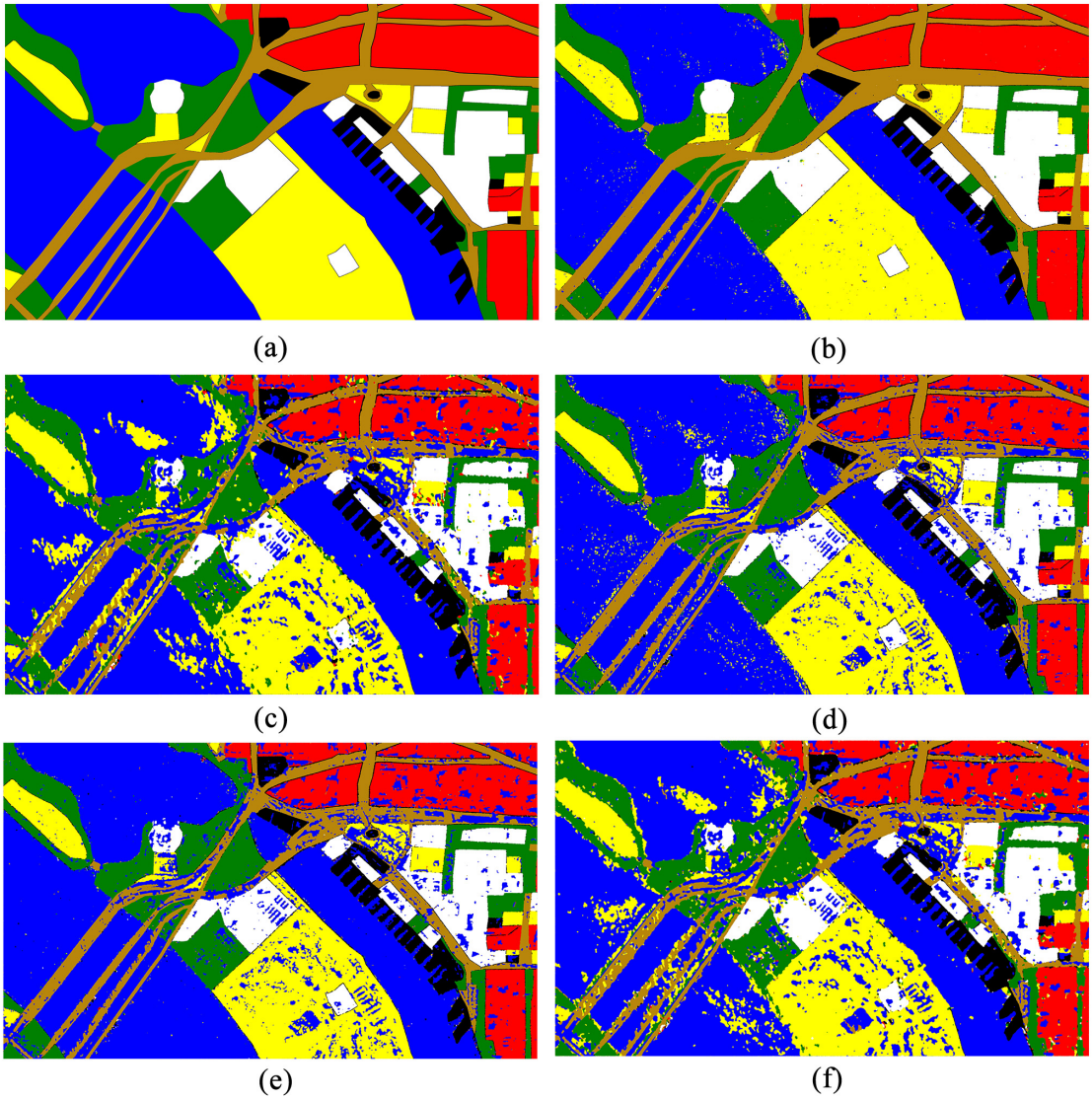



Fig. 14. Classification maps of each method on Washington SAR image. (a) Ground truth. (b) MCAR-CAN. (c) CNN. (d) DPN. (e) NasNet. (f) ResNet.

and the comparison algorithms [Fig. 13(c)–(f)] for the Rosenheim SAR image. By comparing the experimental results with the ground truth, it is apparent that the MCAR-CAN exhibited a better classification effect. Our approach captured more detailed multiscale information and contextual information to describe the regional boundaries. The classification results of the four comparison algorithms exhibited some errors, i.e., the classification results of trees and farmland were confused with each other. Complex geographical conditions reduced the classification accuracy of these four comparison methods. From this SAR image, our proposed algorithm as well as the four comparison algorithms could all extract a pure water surface area. Although all of the algorithms could roughly estimate the distribution of the various terrains in these experiments, our proposed algorithm proved to be more accurate and better.

Fig. 14 is a visualization of the experimental results for the Washington SAR image. The resolution of this SAR image is higher, and the different terrains are more detailed. In this image, buildings are subdivided into high-rise and low-rise, a distinction not made in the previous two SAR images. Therefore, higher resolution and more detailed terrain increase the classification difficulty. The experimental result of the CNN [Fig. 14(c)] showed that a large number of water pixels were misclassified as either trees or bare land. In addition, roads and buildings were incorrectly classified as water. The experimental result of DPN [Fig. 14(d)] was better than that of the CNN. The number of pixels that were misclassified was greatly reduced in water area and bare land. However, roads and buildings are still mistakenly classified as water. In Fig. 14(e), it can be seen that the classification performance of NasNet is higher than that of DPN. But the result of NasNet was still unsatisfactory. Lots of buildings, roads, and land were misclassified as water. The ResNet [Fig. 14(f)] still generated experimental results similar to those of the CNN. The high-resolution and complex terrain of the Washington SAR image created enormous difficulties for each algorithm. Surprisingly, our proposed algorithm still maintained good classification ability. The MCAR-CAN not only could clearly

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TABLE II
ARCHITECTURE OF COMPARISON METHODS

| output size | CNN | DPN | NasNet | ResNet |
|---|---|---|---|---|
| 25 × 25 | 3 × 3, 16<br>3 × 3, 16 | 3 × 3, 64 | 3 × 3, 64<br>Normal Cell × 4 | 3 × 3, 32 |
| 13 × 13 | 3 × 3, 32, stride 2<br>3 × 3, 23 | 3 × 3 max pool, stride 2<br>$\begin{bmatrix} 1\times1, 96 \\ 3\times3, 96, G=32 \\ 1\times1, 256\ (+16) \end{bmatrix} \times 3$ | Reduction Cell<br>Normal Cell × 4 | $\begin{bmatrix} 3\times3, 32 \\ 3\times3, 32 \end{bmatrix} \times 2$ |
| 7 × 7 | 3 × 3, 64, stride 2<br>3 × 3, 64 | $\begin{bmatrix} 1\times1, 192 \\ 3\times3, 192, G=32 \\ 1\times1, 512\ (+32) \end{bmatrix} \times 4$ | Reduction Cell<br>Normal Cell × 4 | $\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix} \times 2$ |
| 4 × 4 | 3 × 3, 128, stride 2 | $\begin{bmatrix} 1\times1, 384 \\ 3\times3, 384, G=32 \\ 1\times1, 1024\ (+32) \end{bmatrix} \times 4$ | Reduction Cell<br>Normal Cell × 4 | $\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix} \times 2$ |
| 1 × 1 | 2048-d fc<br>512-d fc<br>256-d fc<br>n-d, softmax | average pool<br>n-d fc, softmax | average pool<br>n-d fc, softmax | average pool<br>n-d fc, softmax |

Convolutional layers are without special instructions. In the third column, DPN unit is in brackets. In the fifth column, ResNet unit in brackets.

TABLE III
OA OF CLASSIFICATION RESULTS ON THREE SAR IMAGES (%)

| Method | Image 1 | Image 2 | Image 3 |
|---|---|---|---|
| MCAR-CAN | **98.14** | **97.05** | **97.09** |
| CNN | 91.04 | 92.46 | 80.18 |
| DPN | 95.63 | 94.57 | 89.12 |
| NasNet | 92.27 | 95.48 | 90.37 |
| ResNet | 94.26 | 91.85 | 81.05 |

TABLE IV
AA OF CLASSIFICATION RESULTS ON THREE SAR IMAGES (%)

| Method | Image 1 | Image 2 | Image 3 |
|---|---|---|---|
| MCAR-CAN | **98.68** | **95.06** | **96.98** |
| CNN | 93.09 | 91.54 | 78.16 |
| DPN | 97.03 | 93.86 | 87.44 |
| NasNet | 94.72 | 95.03 | 88.16 |
| ResNet | 95.75 | 91.57 | 79.50 |

TABLE V
KAPPA OF CLASSIFICATION RESULTS ON THREE SAR IMAGES

| Method | Image 1 | Image 2 | Image 3 |
|---|---|---|---|
| MCAR-CAN | **0.9812** | **0.9689** | **0.9651** |
| CNN | 0.9102 | 0.9245 | 0.8003 |
| DPN | 0.9562 | 0.9457 | 0.8910 |
| NasNet | 0.9226 | 0.9548 | 0.9036 |
| ResNet | 0.9425 | 0.9184 | 0.8091 |

extract road areas, but could also complete the classification of each area. It is encouraging that the proposed algorithm exhibited good classification performance and strong robustness with different SAR data.

In order to quantitatively assess the experimental results, we calculated the three evaluation indicators—OA (Table III), AA (Table IV), and kappa (Table V). In the three SAR images, the OA of the MCAR-CAN was greater than 97%, which is much higher than the other algorithms. For the different SAR images, the AA of the MCAR-CAN was greater than 95%, and the kappa coefficient was higher than 0.96.

TABLE VI
OA OF CLASSIFICATION RESULTS ON ROSENHEIM SAR IMAGE WITH 2%–10% TRAINING SAMPLES (%)

| Method | 2% data | 4% data | 6% data | 8% data | 10%data |
|---|---|---|---|---|---|
| MCAR-CAN | **83.32** | **87.44** | **90.27** | **93.03** | **97.05** |
| CNN | 78.90 | 82.78 | 86.28 | 89.38 | 92.46 |
| DPN | 75.21 | 81.93 | 87.59 | 91.01 | 94.57 |
| NasNet | 76.33 | 82.65 | 87.20 | 91.22 | 95.48 |
| ResNet | 79.37 | 83.24 | 86.54 | 89.01 | 91.85 |

Tables III–V demonstrate that the proposed algorithm has excellent classification ability. For the first two SAR images, in our experiments, the OA and AA of four comparison algorithms were above 91% and the kappa coefficient was higher than 0.91. And the results of the four comparison algorithms were very deficient in the Image 3 experiment. OA and AA were lower than 91%, and kappa was less than 0.91. The four comparison algorithms were not particularly robust for different SAR data. In contrast, the MCAR-CAN still produced good experimental results. It is clear that the proposed method is more robust and accurate than the other algorithms.

We also conducted comparative experiments that confirmed the classification ability of these algorithms with different volumes of data in the Rosenheim SAR image. Between 2% and 10% of data was randomly extracted as training data to train the model. Table VI lists the OA of experimental results. It is well known that the more training data there are, the better the features are learned. It can be concluded from Table VI that the classification accuracy of all of the algorithms decreases as the amount of data decreases from 10% to 2%. The accuracy of the MCAR-CAN, CNN, DPN, NasNet, and ResNet decreased by 13.73%, 13.56%, 19.36%, 19.15%, and 12.48%, respectively. The classification accuracy of the MCAR-CAN is much higher than that of other algorithms under different

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data volumes. And, the MCAR-CAN has greater robustness and learning ability.

## IV. Conclusion

In this study, the new MCAR-CAN algorithm is proposed for SAR image classification. The MCAR-CAN has two branches. In the autoencoder branch, asymmetric structure can make the network more focused on learning classification information since the decoder regularizes the encoder. Inspired by the attention mechanism, robust and effective features are obtained by producing an attention weight corresponding to context correlation in the attention branch. Latent representations from the two branches are combined and input into the classifier for training. The experimental results for three SAR images prove the superiority of our proposed algorithm over other algorithms. In addition, experimental results using different data volumes prove that our algorithm has better robustness and learning ability. The mixed training strategy and postprocessing algorithm improve the classification accuracy to some extent.

Our study has demonstrated the efficacy of autoencoder regularization and the attention mechanism for SAR image classification. In future work, we plan to generalize our method for more complex remote sensing applications.

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